

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science & Engineering (CSE)
1st Year 1st Semester Final Examination 2024
EEE-1103: Electrical Circuits

Time: 3.00 hours

Full Marks: 70

[Answer any 5 (Five) of the Following Questions]

1. (a) Write down Kirchhoff's laws. 02
- (b) Draw the Symbol of different types of dependent sources. 02
- (c) The current through an element is shown in Fig. 1.1. Determine the total charge that passed through the element at: 05

(a) $t = 1$ s (b) $t = 3$ s (c) $t = 5$ s

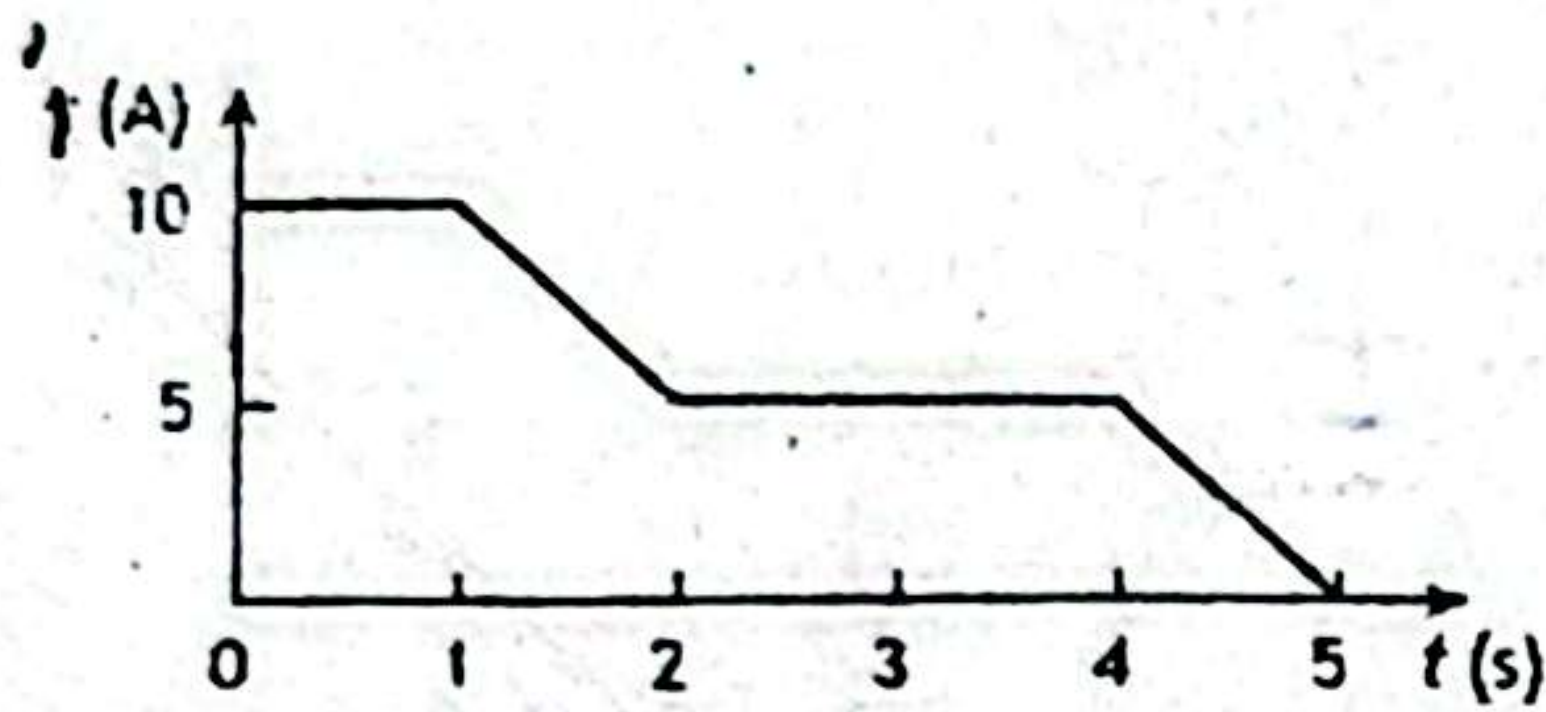


Fig: 1.1

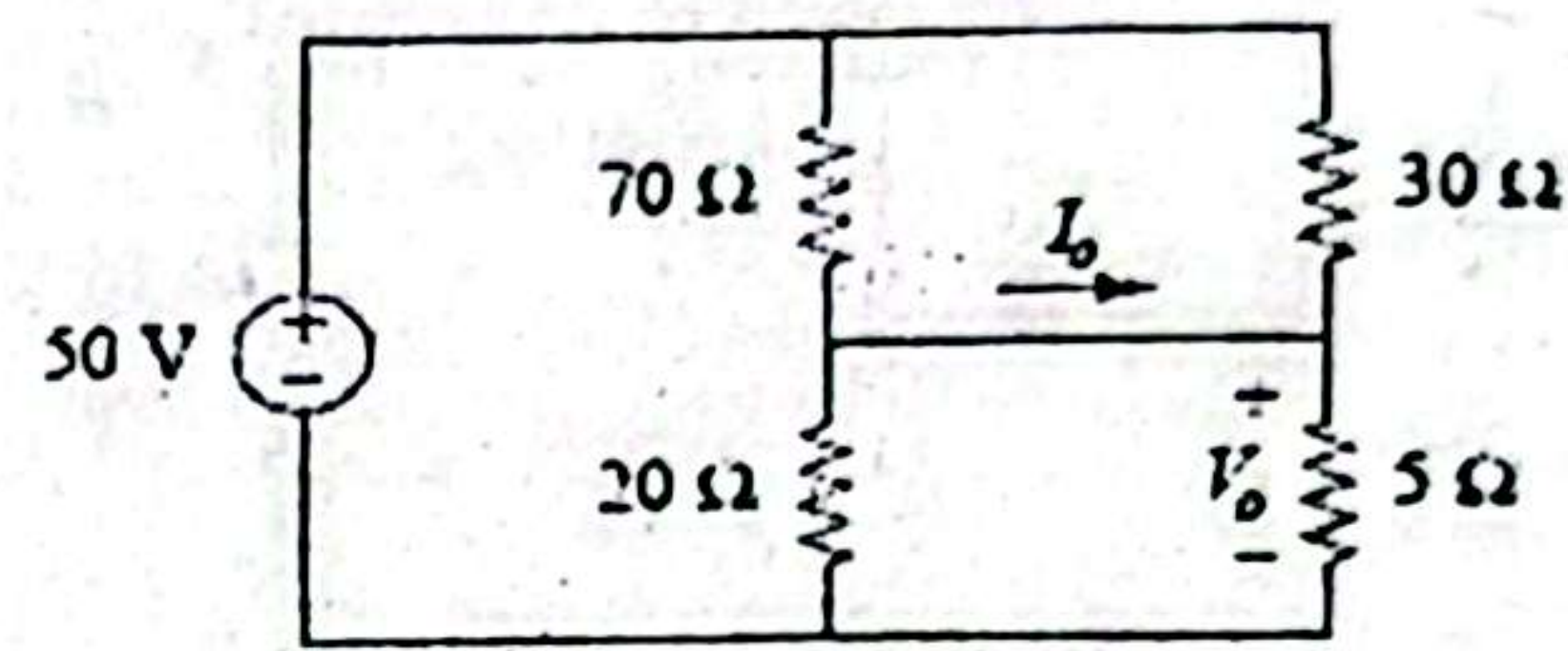


Fig: 1.2

- (d) Calculate V_0 and I_0 of the circuit in Fig 1.2 05
2. (a) For Wye-Delta conversion, Prove $R_\Delta = 3R_Y$ 04
- (b) For the bridge network in Fig. 2.1, find R_{ab} and i . 05
- (c) Find node voltages using nodal analysis of the following circuit in Fig 2.2. 05

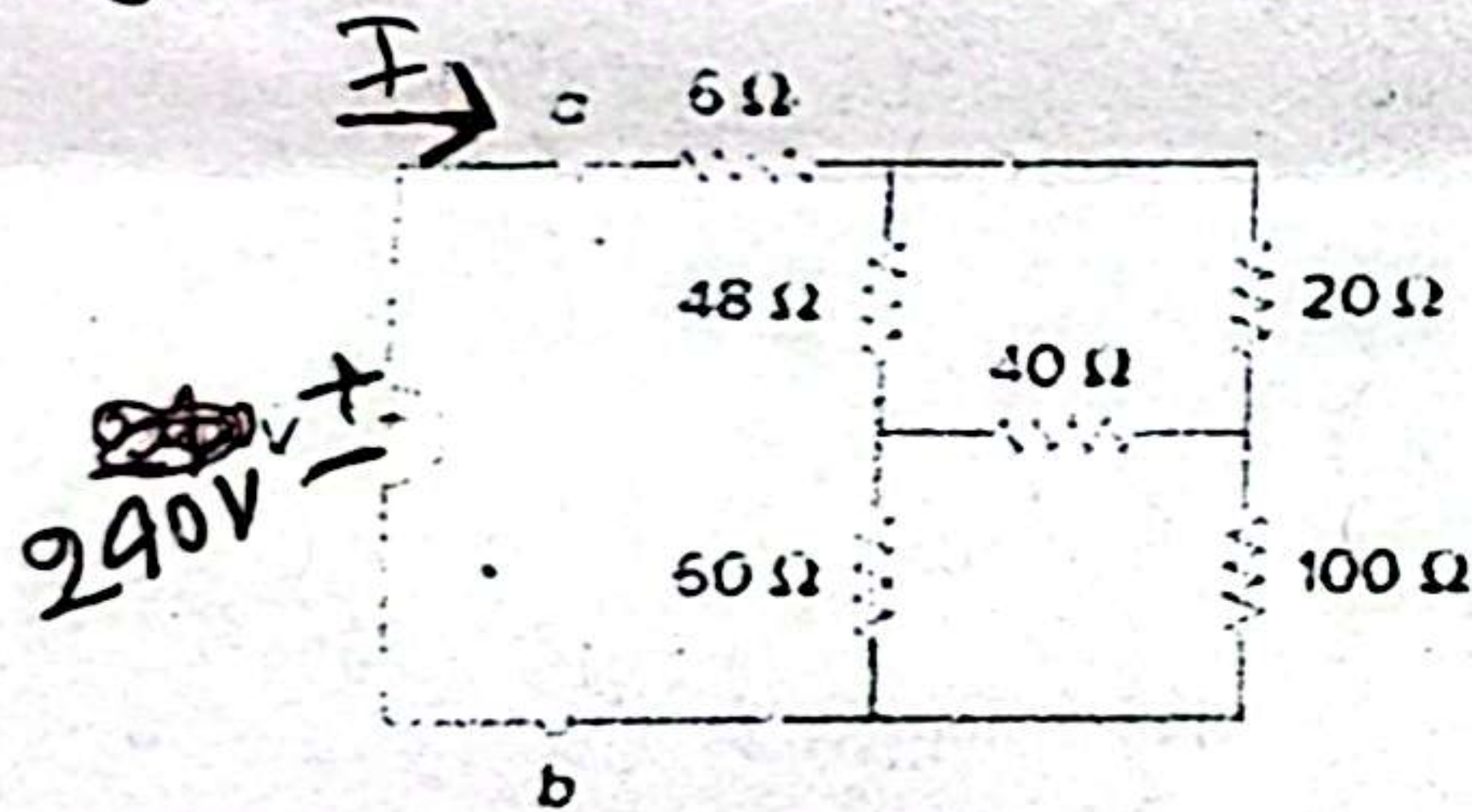


Fig: 2.1

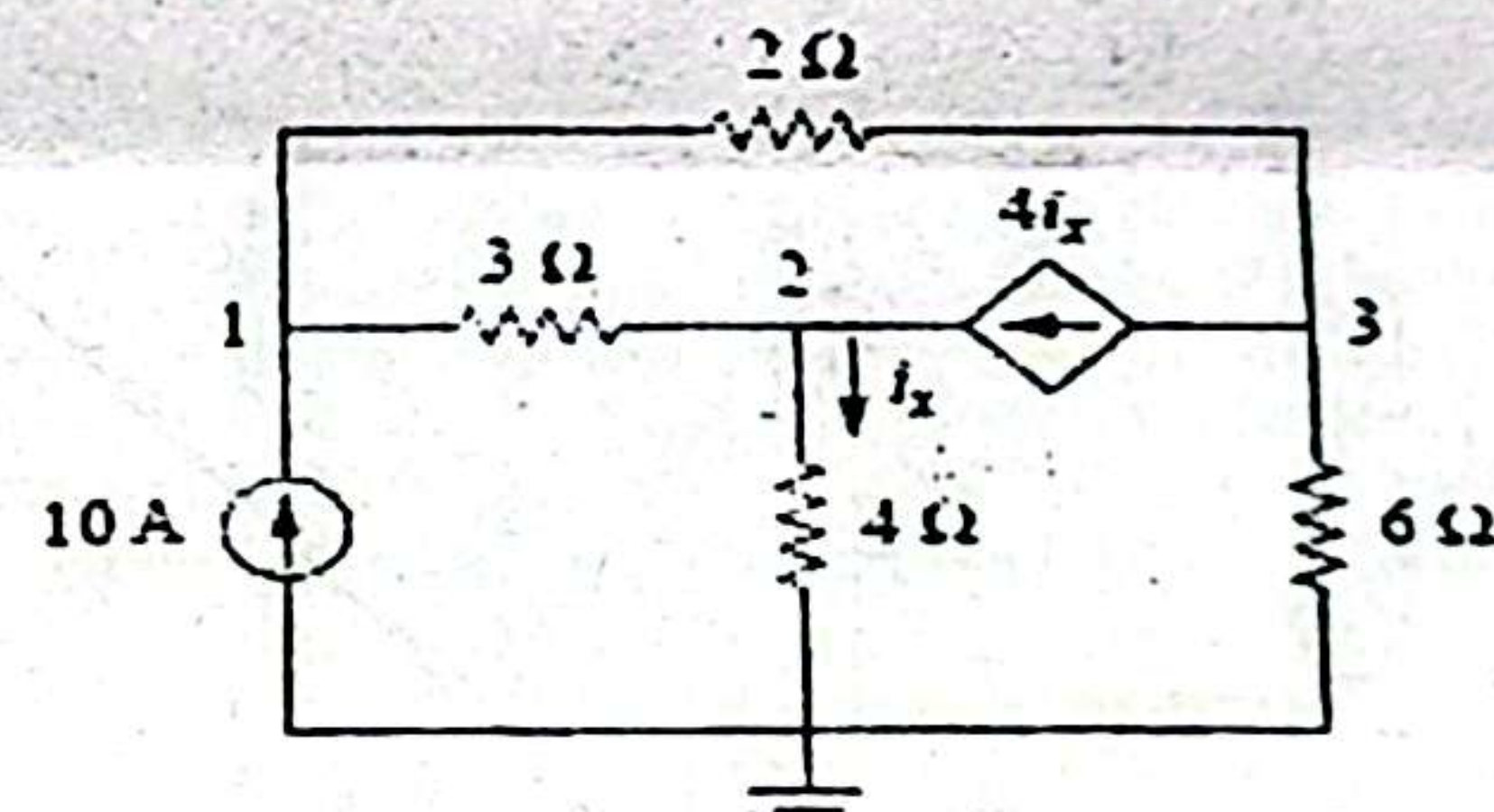


Fig: 2.2

3. (a) Deduce the expression of the Voltage Divider Rule (VDR) with a suitable figure. 04
- b) Obtain v_1 through v_3 in the circuit of Fig-3.1. 04
- (c) Use the mesh analysis to find the currents i_1, i_2, i_3 and v_0 in the network of Fig-3.2. 06

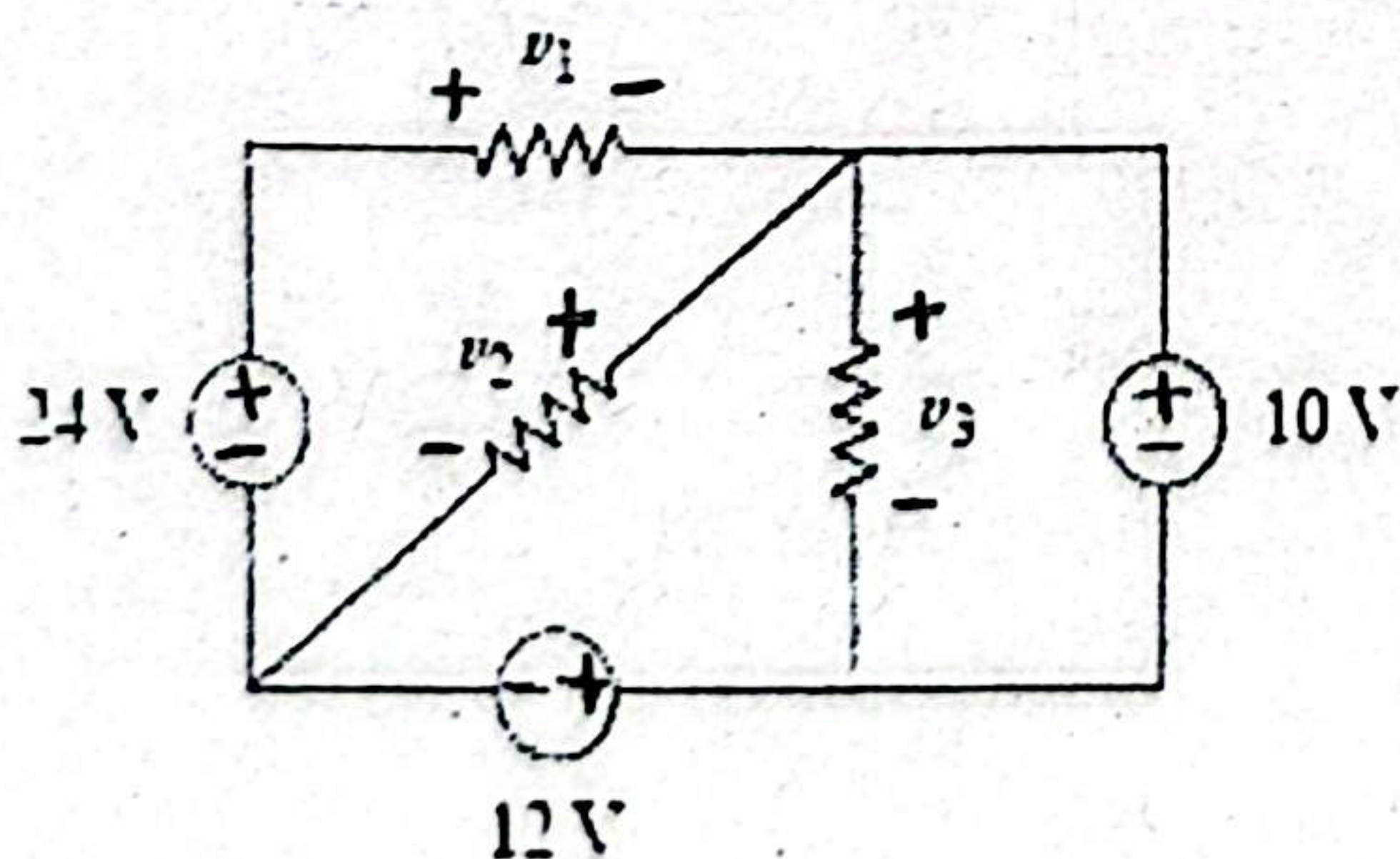


Fig 3.1

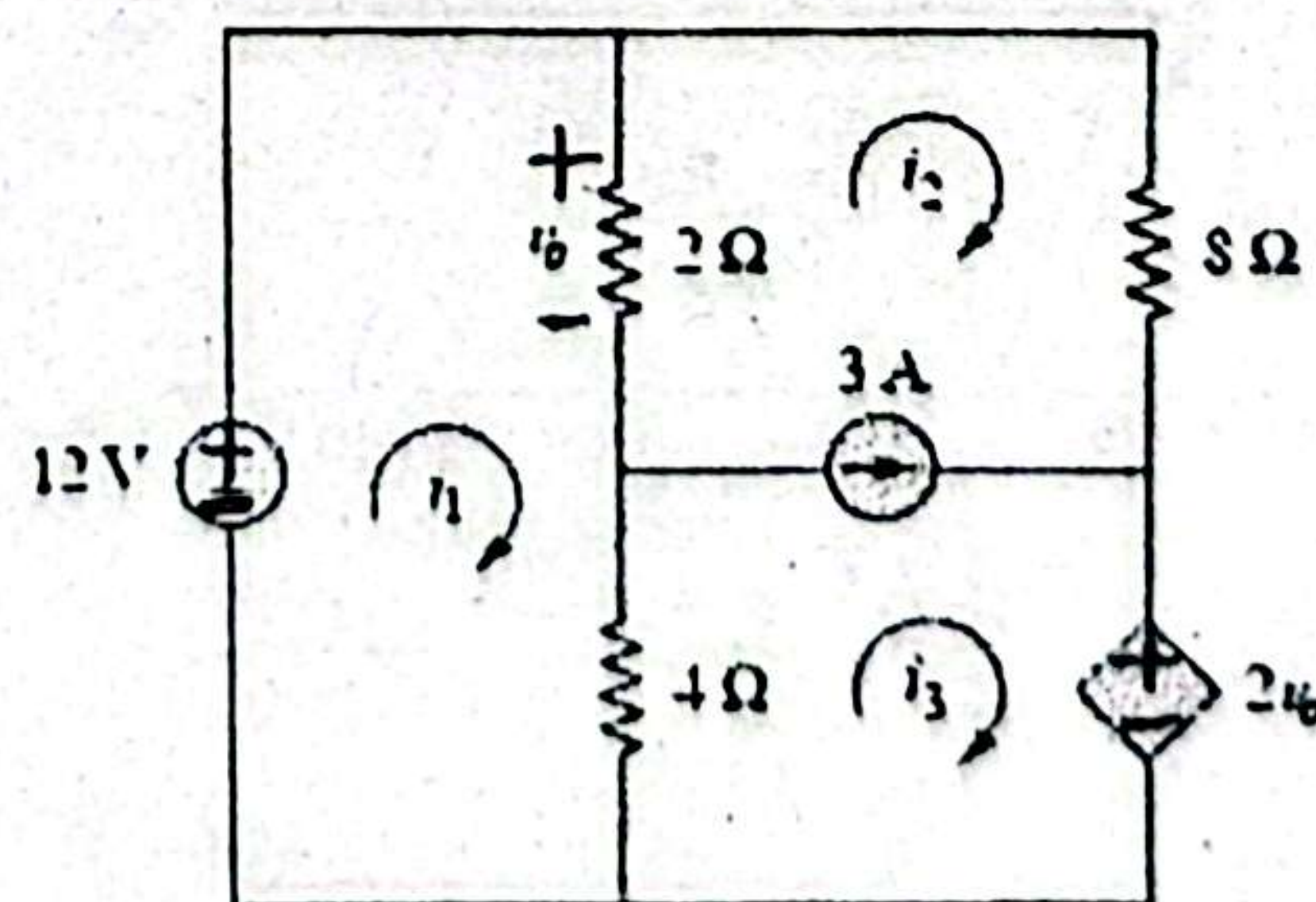


Fig 3.2

4. (a) State Source Transformation Theorem. 02
- b) Find Norton's equivalent circuit of the following circuit in Fig 4.1. 06
- (c) Find v_0 of the circuit in Fig 4.2 using source transformation. 06

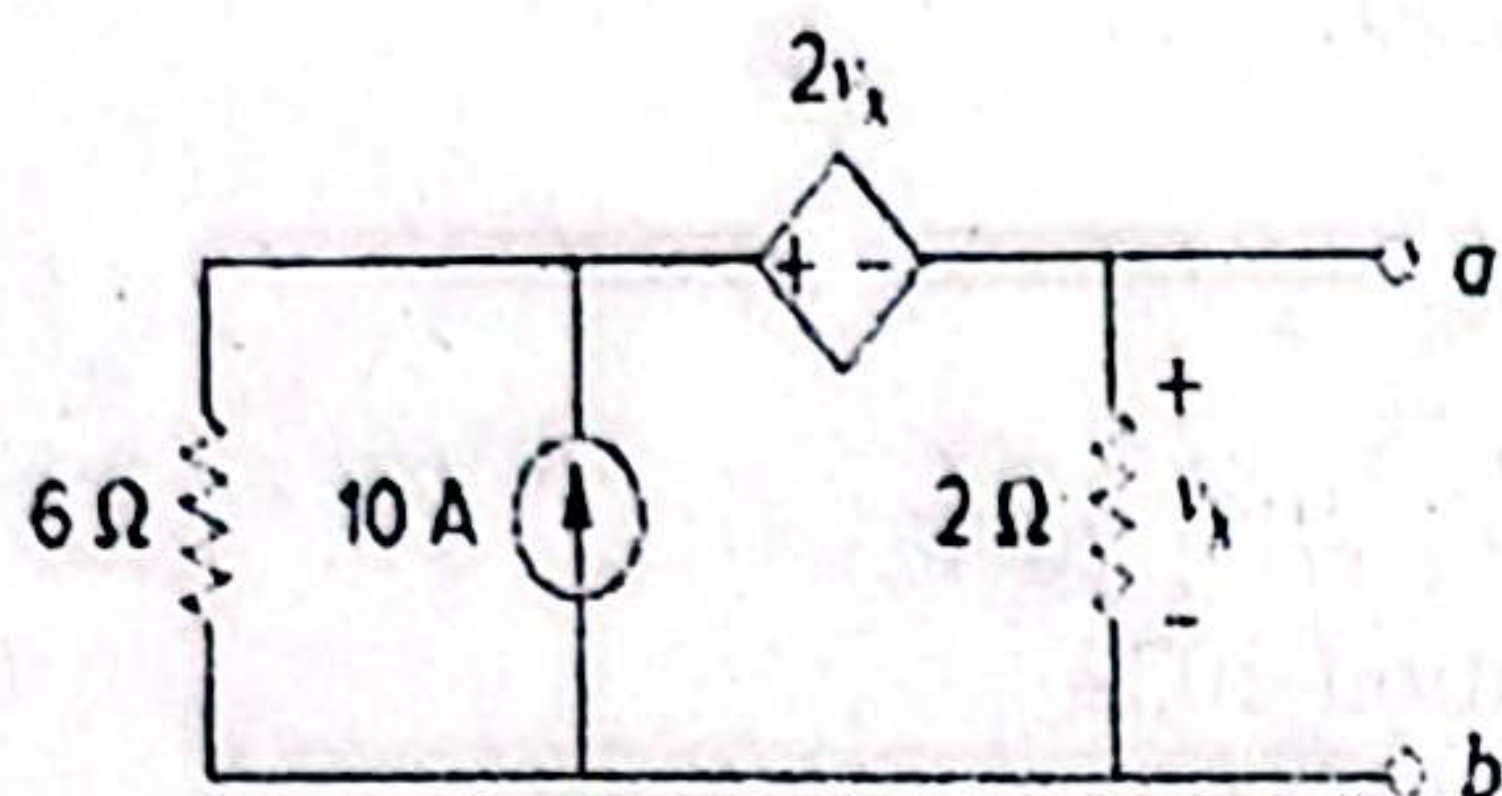


Fig: 4.1

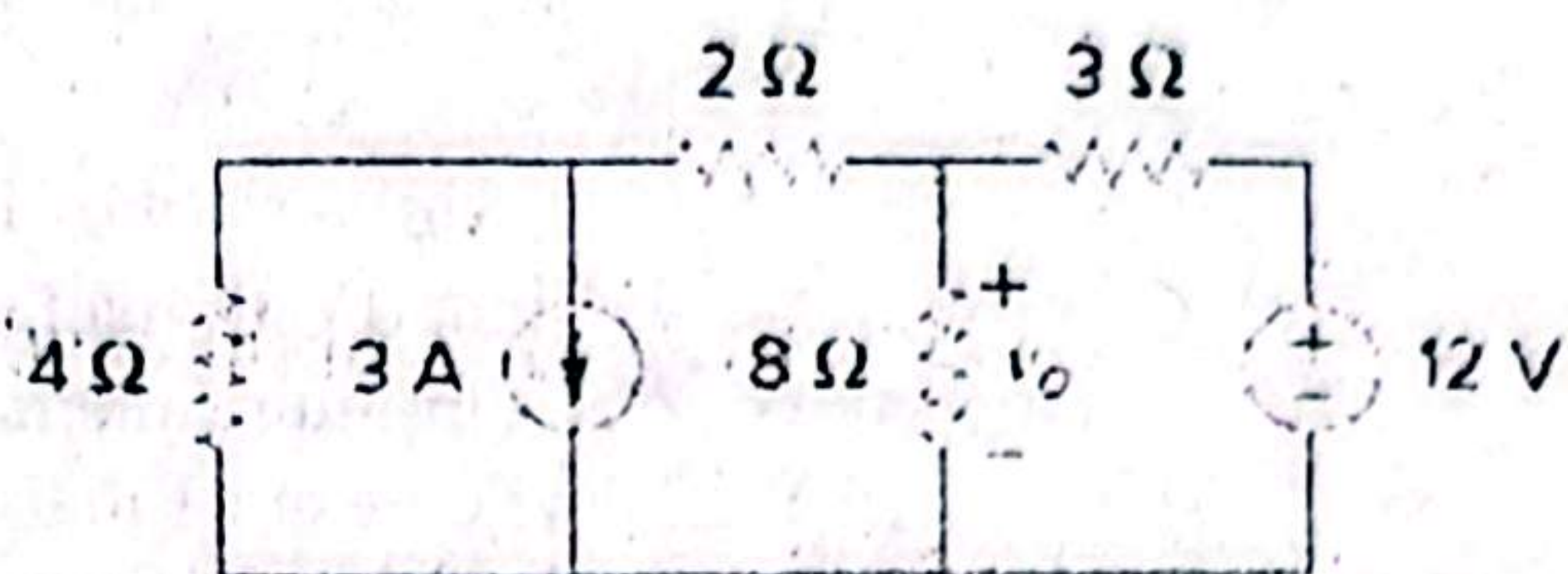


Fig: 4.2

5. (a) For the maximum power transfer theorem prove that maximum power will be transfer to the load when $R_L = R_{Th}$. 04
- (b) Use the Superposition theorem to find current i in the circuit of Fig-5.1. 06
- c) Find the Thevenin equivalent circuit in the network of Fig-5.2. 04

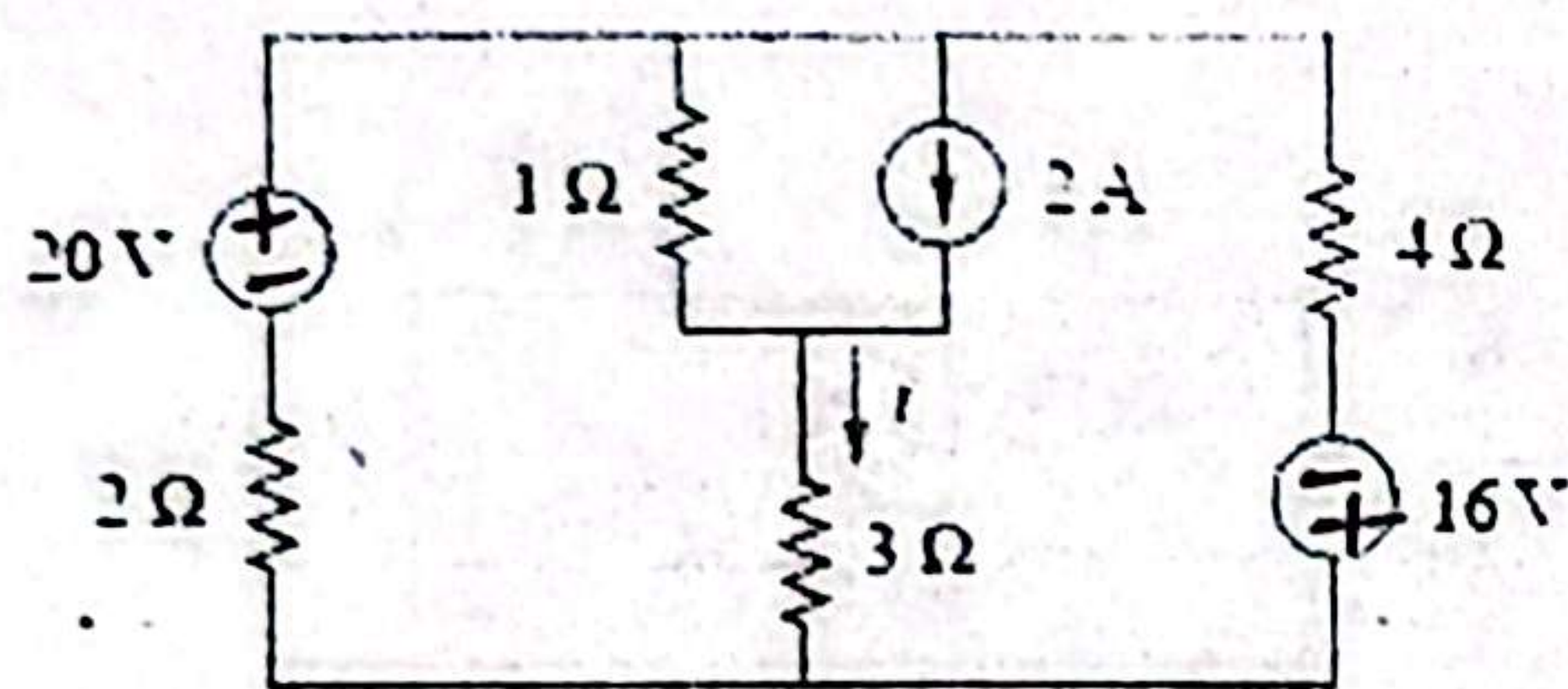


Fig: 5.1

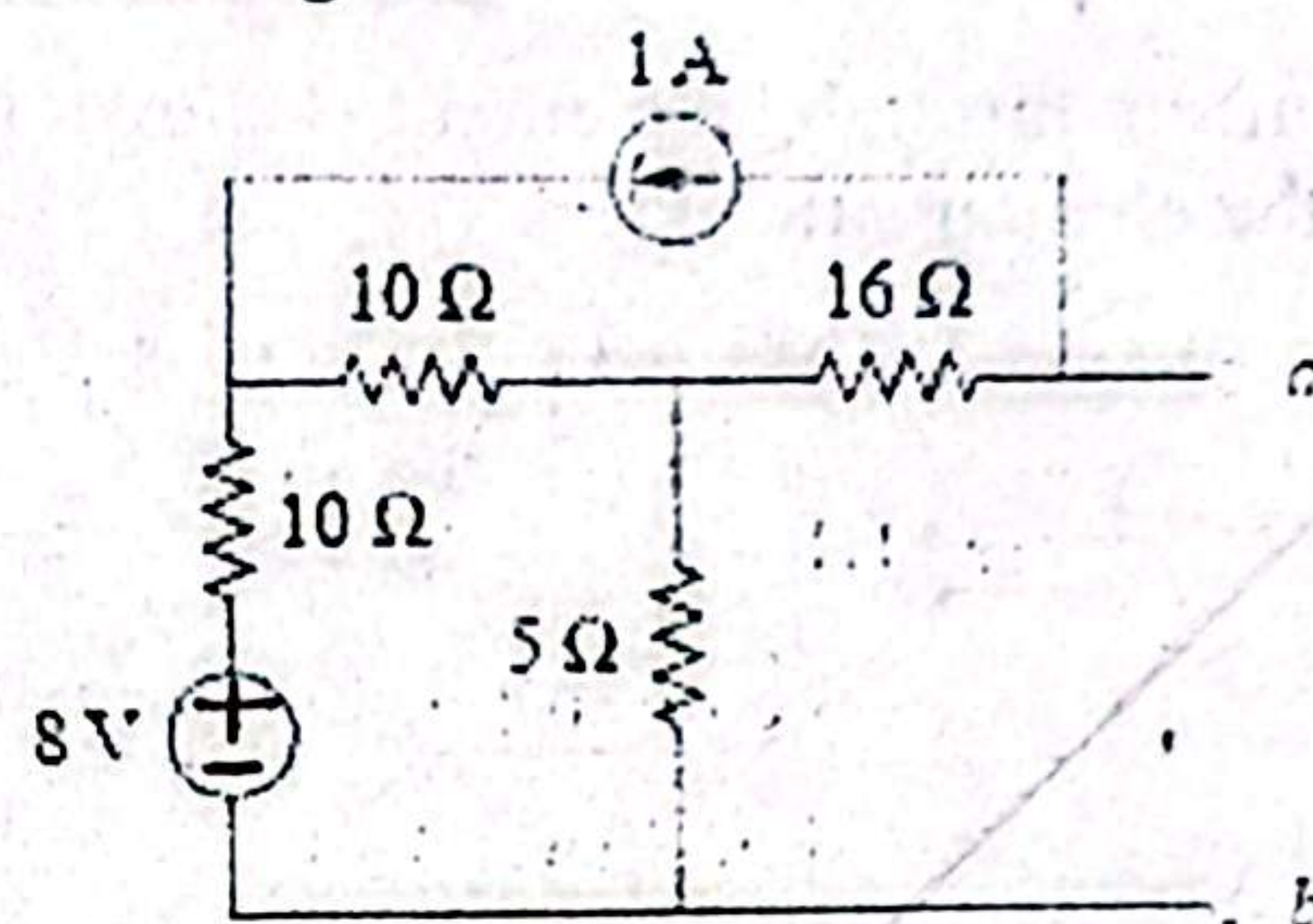


Fig: 5.2

- a) Show that the equivalent capacitance of N series connected capacitors is the reciprocal of the sum reciprocals of the individual capacitances. 04
- b) Derive the equation for the energy stored in an inductor. 04
- c) Determine the current i_L in the network of Fig-6.1. 03
- d) Find the equivalent capacitance for the circuit in the Fig-6.2. 03

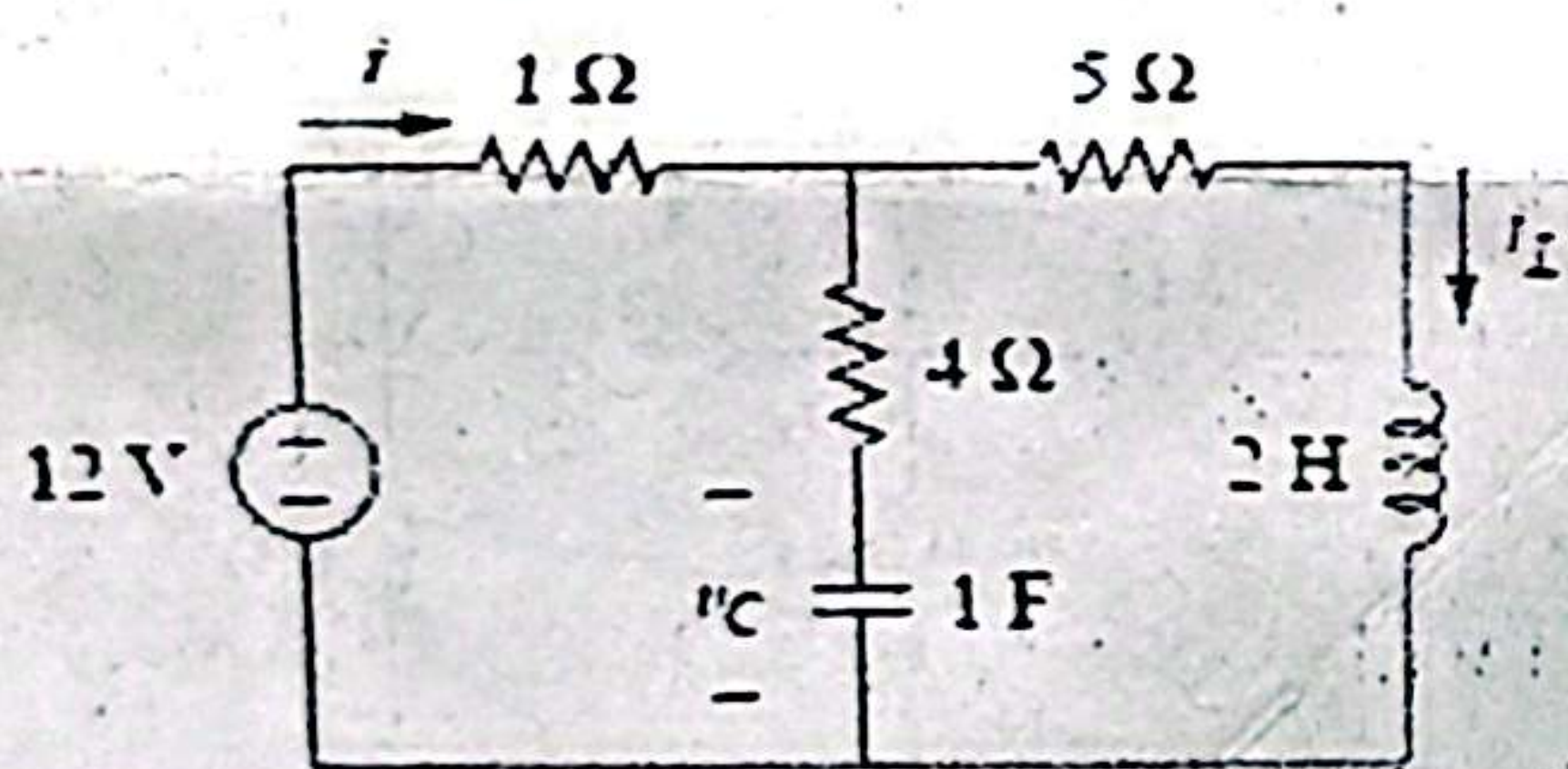


Fig: 6.1

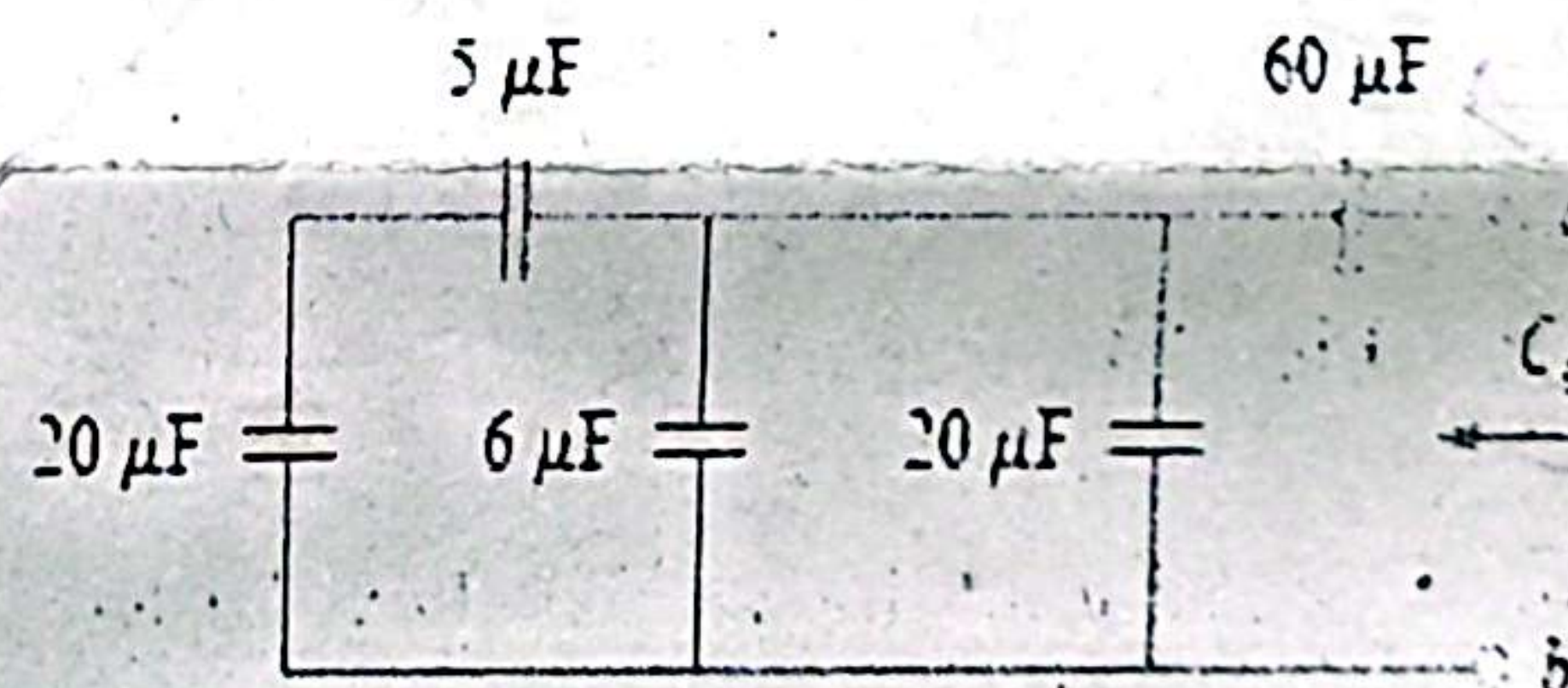


Fig: 6.2

- a) Derive the equation of Effective power absorbed by a resistor. 03
- b) Determine the rms value of the current waveform of the following Fig 7.1. If the current passes through a 8-Ω resistor, find the average power absorbed by the resistor. 05
- c) Determine the average power generated by each source and average power absorbed by each passive element of the following circuit in Fig 7.2 06

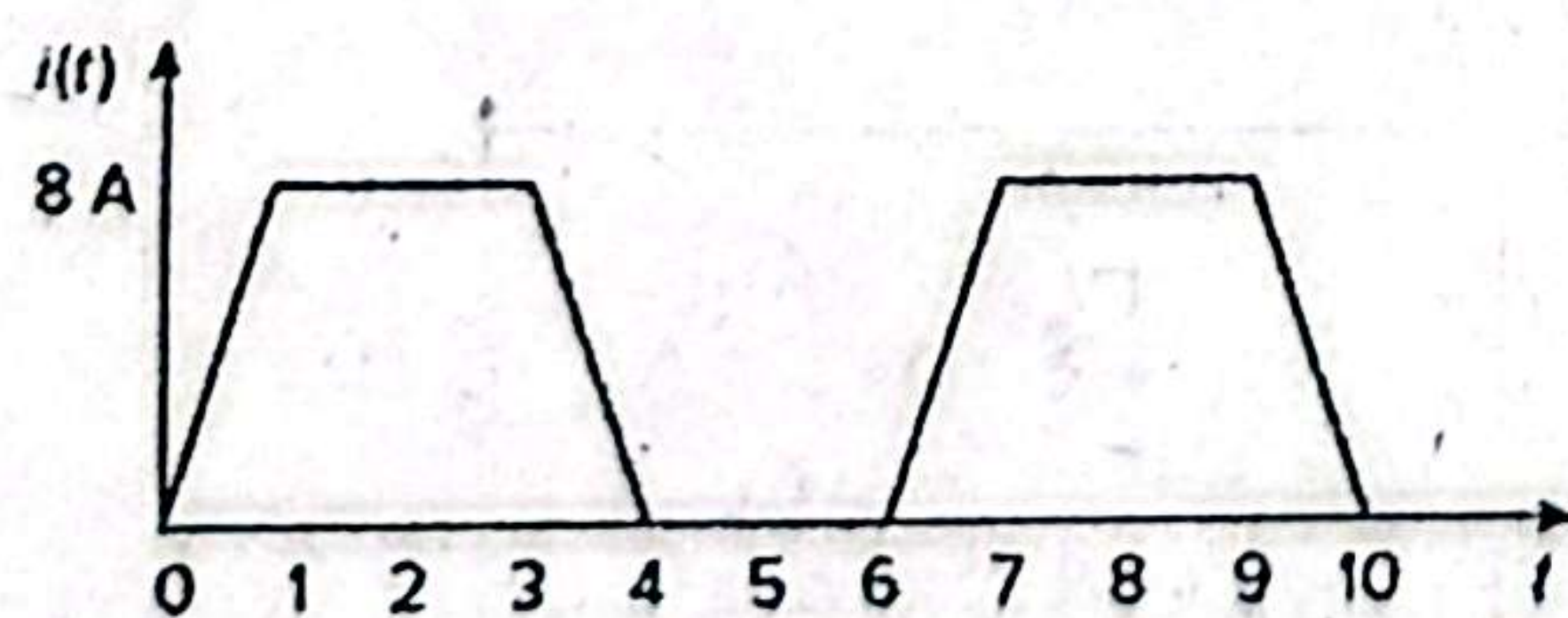


Fig: 7.1

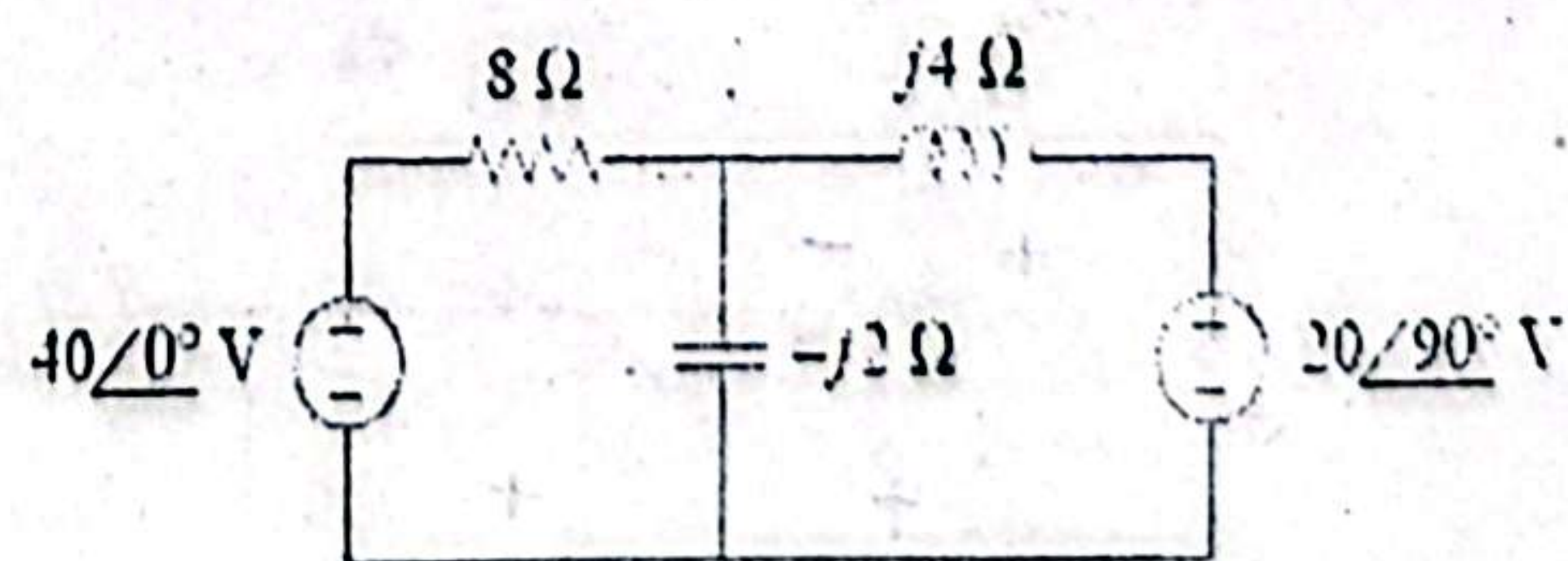


Fig: 7.2